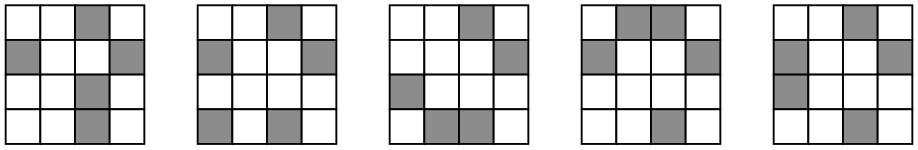
Preview: set-01/by_level/Grade_03-04_Ecolier/y2023_q24 (v1)

One guess is right and each other guess has all but one right: test every option against all the others and count what they have in common

How to think about it: one guess is right, the others have one mistake

If one guess is right, then every other guess must have exactly 4 of its 5 cells in common with the right one. So test each pattern: **count the grey cells it shares with every other pattern**.



The contest question

Count the cells in common for each pair (row = the pattern you test):

	A	B	C	D	E
A	-	4	3	4	4
B	4	-	3	4	4
C	3	3	-	3	4
D	4	4	3	-	4
E	4	4	4	4	-

Only row **E** has a 4 everywhere. So Maria coloured pattern E.

A quicker way

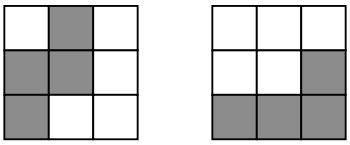
Count how many patterns colour each cell. The right pattern keeps the popular cells: every wrong guess moved just one cell away from it.

Watch out

- Do not stop after comparing two patterns: A and B also share 4 cells, but A is not the answer.

Warm-up: small steps first (no points)

W1. How many grey cells do these two patterns have in common (grey in both)?



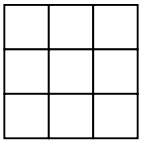
A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 4 ***E.*** 7

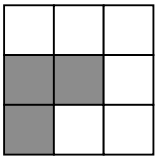
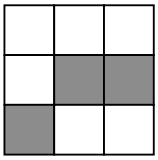
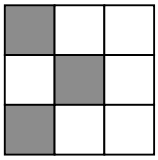
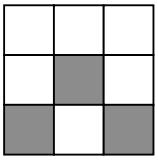
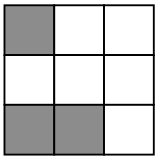
W2. Two friends guess a 4-digit code: 9566 and 9586. In how many places do the two guesses have the same digit?

A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

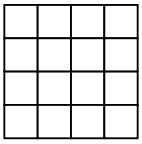
Part A: Each question is worth 3 points

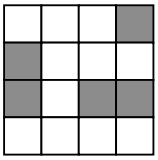
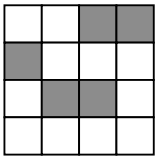
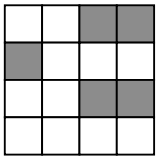
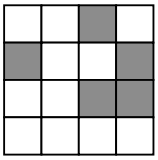
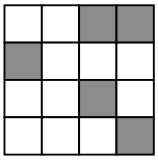
1. Sara colours exactly 3 cells of a 3 × 3 grid grey. Five friends guess which cells Sara coloured: the patterns A to E. Sara says: "One of you is right. Each of the others guessed exactly 2 cells correctly." Which pattern did Sara colour?



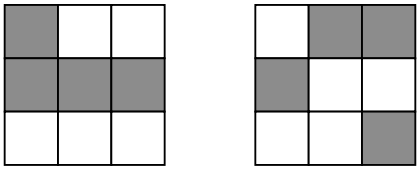
A.  ***B.***  ***C.***  ***D.***  ***E.*** 

2. Ben colours exactly 5 cells of a 4 × 4 grid grey. Five friends guess which cells Ben coloured: the patterns A to E. Ben says: "One of you is right. Each of the others guessed exactly 4 cells correctly." Which pattern did Ben colour?



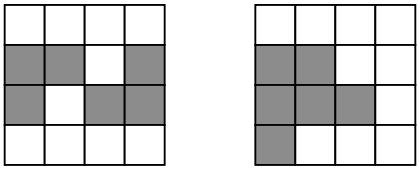
A.  ***B.***  ***C.***  ***D.***  ***E.*** 

3. Ben coloured the left grid. A friend guessed the right grid. How many cells did the friend guess correctly?



A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 6

4. Sara coloured the left grid. A friend guessed the right grid. How many cells did the friend guess correctly?



A. 4 ***B.*** 5 ***C.*** 6 ***D.*** 8 ***E.*** 9

Part B: Each question is worth 4 points

5. Mia sets a secret code with 4 digits on a bike lock. Five friends guess the code: 2366, 7366, 7316, 7369 and 7666. Mia says: "One of you is right. Each of the others has exactly 3 digits right, in the right place." What is the code?

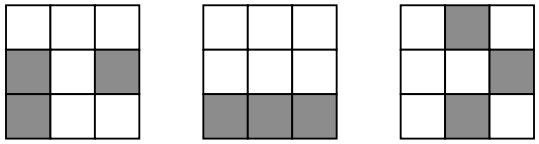
A. 7316 ***B.*** 2366 ***C.*** 7666 ***D.*** 7369 ***E.*** 7366

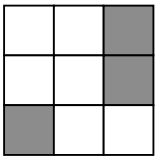
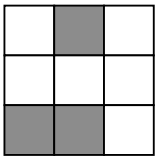
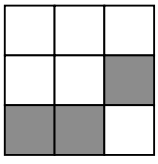
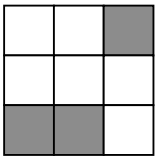
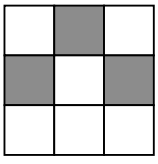
6. Sara sets a secret code with 5 digits on a bike lock. Five friends guess the code: 37388, 30348, 30385, 30380 and 30388. Sara says: "One of you is right. Each of the others has exactly 4 digits right, in the right place." What is the code?

A. 37388 ***B.*** 30385 ***C.*** 30348 ***D.*** 30380 ***E.*** 30388

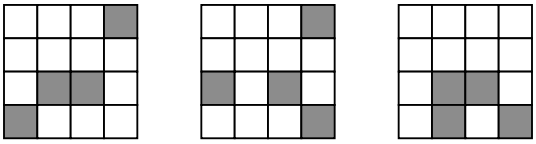
Part C: Each question is worth 5 points

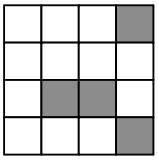
7. Sara colours exactly 3 cells of a 3 × 3 grid grey. Three friends guess (top row). Sara says: "Nobody is right, but each of you guessed exactly 2 cells correctly." Which of the patterns A to E did Sara colour?

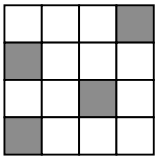


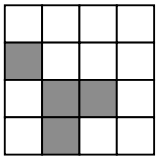
A.  ***B.***  ***C.***  ***D.***  ***E.*** 

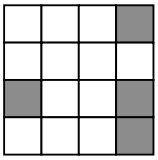
8. Lena colours exactly 4 cells of a 4 × 4 grid grey. Three friends guess (top row). Lena says: "Nobody is right, but each of you guessed exactly 3 cells correctly." Which of the patterns A to E did Lena colour?

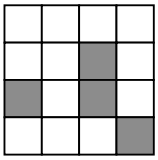


A. 

B. 

C. 

D. 

E. 

Answer Key

W1.	**B**	**W2.**	**C**	**1.**	**C**	**2.**	**C**
5.	**E**	**6.**	**E**	**7.**	**C**	**8.**	**A**

3.	**A**	**4.**	**A**

Solutions

W1. B (1) — Grey in both: 1. [set-01/by_level/Grade_03-04_Ecolier/2023 Q24 · w_common]

W2. C (3) — Compare digit by digit: 3 places are the same. [set-01/by_level/Grade_03-04_Ecolier/2023 Q24 · w_codes]

1. C — Compare each pattern with all the others. Pattern C has exactly 2 grey cells in common with every other pattern, so it is the right one. [set-01/by_level/Grade_03-04_Ecolier/2023 Q24 · which_pattern]

2. C — Compare each pattern with all the others. Pattern C has exactly 4 grey cells in common with every other pattern, so it is the right one. [set-01/by_level/Grade_03-04_Ecolier/2023 Q24 · which_pattern]

3. A (1) — Look for cells that are grey in both grids: 1. [set-01/by_level/Grade_03-04_Ecolier/2023 Q24 · how_many_right]

4. A (4) — Look for cells that are grey in both grids: 4. [set-01/by_level/Grade_03-04_Ecolier/2023 Q24 · how_many_right]

5. E (7366) — Compare each guess with all the others, digit by digit. Only 7366 has exactly 3 digits in the same places as every other guess. [set-01/by_level/Grade_03-04_Ecolier/2023 Q24 · lock_code]

6. E (30388) — Compare each guess with all the others, digit by digit. Only 30388 has exactly 4 digits in the same places as every other guess. [set-01/by_level/Grade_03-04_Ecolier/2023 Q24 · lock_code]

7. C — Check each pattern against all three guesses: pattern C has exactly 2 cells in common with each of them. [set-01/by_level/Grade_03-04_Ecolier/2023 Q24 · not_a_guess]

8. A — Check each pattern against all three guesses: pattern A has exactly 3 cells in common with each of them. [set-01/by_level/Grade_03-04_Ecolier/2023 Q24 · not_a_guess]